

Experimental Design, Participant Selection and Eligibility

Supporting documentation for the replication package.

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1. Experimental Design

To test the spiral of silence hypothesis, we conduct an experiment with UC Berkeley undergraduate students. We first solicit participants' private views on a set of socially sensitive topics and then provide all participants with an opportunity to express their views publicly. Capturing both private and public opinion allows us to determine the prevalence of self-censorship and whether socially appropriate views are expressed more often in public. To test whether attention to silence affects beliefs about others and public expression over time, we experimentally vary the salience of silence when sharing information about discussions by previous groups. We then assign participants to public discussion groups with other participants who received the same information treatment to test the dynamic, self-reinforcing effects.

1.1 Study Procedures

Baseline Survey

All participants first complete a baseline survey about their private views on 10 potentially sensitive socioeconomic and political topics. Participants are then asked to guess the views of other students on the same set of topics. Specifically, for private views, we ask participants to choose either "agree" or "disagree" for each statement.

Participants can also choose to skip any question if they do not want to provide their private views. In practice, only 1% of participants skipped questions about their private beliefs in the baseline survey.

For guesses about the views of other students, we ask *"Among all other Berkeley students who participate in this study, what percentage do you think will privately answer 'agree' and 'disagree' [with this statement]?"*. Survey responses about the beliefs of other students' views are incentivized for accuracy using a binarized scoring rule.

We also collect demographic information, including gender, ethnicity, school year, major, and self-identified political affiliation.

Group Discussion Sessions

Discussion Procedure: All respondents then participate in Zoom sessions of approximately 8-13 participants, where they have the opportunity to express their views publicly to other students on a subset of topics. To make the Zoom session feel more like public discourse with peers, participants are asked to keep their camera on throughout the session and to display their first name and the first initial of their last name. To receive study compensation, students just need to

be present during the Zoom discussion, but they are not paid based on whether or how often they speak. Each Zoom session is facilitated by a trained moderator using a standardized script.

Within each Zoom session, the same four potentially socially sensitive topics are discussed: Renaming Schools, Affirmative Action, Death Penalty and Immunizations. Each topic is first briefly introduced by the moderator, who then reads a statement on the topic. For example, for the renaming schools topic, the moderator reads the statement “*All public schools named after controversial historical figures, including former Presidents George Washington, Thomas Jefferson, and Abraham Lincoln, should be renamed.*” All participants then have 90 seconds to decide whether they would like to share their views on the topic with other students in the Zoom room. Participants first message the moderator privately indicating whether they “agree” or “disagree” with the statement if they want to share their views publicly. Students who do not wish to share their views simply do not message the moderator. The moderator then calls upon all of the participants who volunteered in a random order to share their views. All participants who indicate they want to share their views are called upon to do so. To shut down potential dynamic effects within each session, the order in which participants are called upon is random, and participants speak only when called.

First Movers: A small subset of participants are randomly assigned to be First Movers, who participate in Zoom sessions with other First Movers without any additional information. In total we hosted 4 Zoom Sessions with First Movers, each with 12-13 participants.

Second Movers: We then share a summary (anonymized) of the views expressed publicly by the First Movers with the remaining participants (Second Movers). Importantly, we randomly vary whether Second Movers receive “Control” or “Treatment” information about the public views of the First Movers. Specifically, Control participants receive information about the share of First Movers who publicly agreed or disagreed with each topic. For example, we inform Control participants that “*25 Berkeley students like you discussed over Zoom if they agreed with the following statement Here is a summary of their Zoom discussion*” along with a pie chart showing 7 “Agree” and 5 “Disagree” (see the left panel of Figure 1). Treatment participants receive the same instructions and information as Control participants plus additional information on the number of First Movers who remained silent on each topic. Using the same example as above, Treatment students see a pie chart showing 7 “Agree”, 5 “Disagree” and 13 “Silent” (see the right panel of Figure 1). Note that both groups are provided with sufficient information (total number of students in the previous sessions) to determine the number of participants who stayed silent. The key difference between the Control and Treatment information is not the amount of information provided but the *salience* of silence.

The information treatment is embedded in a midline survey that is sent to all Second Movers approximately 12 hours before their respective Zoom sessions. To measure the immediate effects of the information treatment on beliefs, we also re-elicited participants’ beliefs about the views of other students on this set of socially sensitive topics.

For the Second Movers’ Zoom sessions, Control participants are only eligible to sign up for Control Zoom sessions and Treatment participants are only eligible to sign up for Treatment Zoom sessions. In other words, students participate in Zoom sessions with only other students who received the same information. This allows us to study dynamics in societies where people pay similar levels of attention to silence. The protocol for the Zoom sessions for all Second

Movers, including Control and Treatment participants, is otherwise identical to that of the First Movers.

In total, we hosted 36 Zoom sessions with Second Movers, 18 Control and 18 Treatment sessions. Each Zoom session was attended by on average 9 participants, with the smallest session containing 8 attendees and the largest session containing 12 attendees. All participants only participated in one Zoom session. The Zoom sessions each lasted 45 minutes and took place within one to two weeks of the participant completing the baseline survey. The day of week and time of day of each Zoom session was balanced across Control and Treatment sessions so that students' scheduling availability would not be a potential confound.

Endline Survey

At the end of each Zoom session, participants complete an endline survey where they again provide their guesses about the distribution of other students' views. This endline survey allows us to measure how participants update their beliefs after participating in public discourse with other Control/Treatment peers respectively. To assess whether attention to silence is a mechanism, we also elicit participants' recollections of their Zoom discussions, including how many students expressed agreement, disagreement, or stayed silent on each topic. We also ask participants to guess how students who stayed silent would have privately agreed or disagreed with each statement to measure beliefs about selection into silence.

1.2 Topics

We selected the set of socially sensitive topics for our experiment based on current events and opinion articles in popular news outlets and on social media as well as from focus groups with separate samples of college students from Prolific.

For our main experiment, we covered the following five topics in Zoom sessions, which included one placebo topic that is not socially sensitive:

- **Renaming Schools:** All public schools named after controversial historical figures, including former Presidents George Washington, Thomas Jefferson and Abraham Lincoln, should be renamed.
- **Affirmative Action:** If Proposition 209 was repealed, universities in the UC system should adopt extensive affirmative action policies that explicitly take into account race in the admission process.
- **Death Penalty:** The U.S. should abolish the death penalty.
- **Immunizations:** Immunizations, such as for Covid and flu, should be required on Berkeley's campus.
- **Daylight Savings Time "DST" (Placebo):** Daylight savings time should be eliminated.

To measure the social appropriateness of each of these statements, we follow Krupka and Weber (2013) and invite Berkeley students to rate the social appropriateness of these statements using simple coordination games. Students are asked to rate whether each statement is "very inappropriate," "somewhat inappropriate," "neither appropriate nor inappropriate," "somewhat appropriate," or "very appropriate" and are incentivized to match their response with the modal response.

We then use these survey responses to determine which direction of each statement is socially appropriate, pooling together “somewhat appropriate” and “very appropriate” and likewise “somewhat inappropriate” and “very inappropriate.”

2. Sampling and Attrition

We recruited our sample through UC Berkeley’s Experimental Social Science Laboratory (Xlab), whose participant pool includes current UC Berkeley students and recent graduates who voluntarily participate in research studies. In total, we recruited 454 Berkeley students over five rounds from late February to mid April in 2023 through Xlab’s online recruitment system SONA. In terms of demographics, 70% of our participants are female, 54% are Asian and 21% are White. The majority of participants are in their junior or senior year and identify themselves as “Liberal” or “Slightly liberal”.

We recruited in rounds to maintain consistency in the time lag between the baseline survey and subsequent Zoom sessions across all participants and to offer participants more scheduling choices when registering for Zoom sessions. In each round, we recruited approximately 100 participants. Specifically, we recruited 95, 101, 63, 88 and 107 participants during the weeks of February 14, February 28, March 7, March 14 and March 28 respectively.

Participants first completed a baseline survey and then signed up for a Zoom discussion session scheduled for the following week. Participants could select from four designated Zoom sessions based on their treatment assignment. All sessions were scheduled between 3-5pm, Mondays through Thursdays, to eliminate potential confounds that might arise due to different availability on various weekdays or times. Once a student participated in one of the study rounds they were ineligible to participate in subsequent rounds.

Among participants recruited in the first round, a subset of students were randomly assigned to the “First Movers” group and the remaining students to either the “Control” or “Treatment” group. Participants recruited in the subsequent rounds were randomly assigned to either the Control or Treatment groups. In each round, an equal number of students were assigned to the Control and Treatment groups. In total, 56 participants were randomly assigned to the First Movers group, 200 to the Control Group, and 198 to the Treatment group respectively. Assignment of participants to First Movers, Control, and Treatment groups was balanced across observable characteristics and baseline views and beliefs.

In terms of attrition, 454 participants completed the baseline survey, including 56 First Movers and 398 Second Movers. Among the 398 Second Movers who completed the baseline survey, 353 (90%) signed up for a Zoom session and 333 (84%) actually attended their scheduled session. All participants who attended Zoom sessions completed the endline survey. In total, our final sample consists of 383 students who completed all components of the study, including 50 first movers and 333 second movers, of which 166 were assigned to the Control group and 167 were assigned to the Treatment group. Attrition was balanced across treatment assignment, observable characteristics, and baseline views and beliefs. Furthermore, students who dropped out in the Treatment versus Control groups do not differ systematically in observed characteristics including demographics, private beliefs, and baseline guesses about the belief distribution.

3. Randomization Procedure

Randomization was carried out in weekly batches, in coordination with the Xlab lab manager. Each week, once that round's participants had completed the baseline survey, their SONA participant IDs were compiled into a spreadsheet. Assignment was then performed on that spreadsheet in Stata: the random-number seed was set to the date on which the randomization was run, half of the batch was drawn at random to form the Treatment group, that list was exported to Excel and merged back against the full roster, and the participants who did not appear on it formed the Control group. The two resulting lists were sent to the lab manager, who used them to manage sign-up, so that students on the Treatment list could see and sign up only for Treatment Zoom sessions and students on the Control list only Control sessions.